

Path to Product: intelliGEN Opposed Free Piston DI 2-stroke platform performance development



DI2S
Direct Injection 2-Stroke Engines
Conference

26–27 September 2024

Graz University of Technology,
Austria

www.di2-stroke-engine-di2s.com

LIBERTINE
LINEAR POWER SYSTEMS

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intelliGEN P1 Cell



The Linear Generator category

- Linear Generators **convert fuel energy into electricity** directly using linear electrical machines that govern compression ratio in real time whilst also generating electrical power.
- Linear Generators are being adopted in the US for EV truck charging, biogas-to-power, micro-grids, logistics and data centres due to **multiple benefits vs. fuel cells and conventional engines**.

Linear Generators can produce clean, on-demand power from renewable fuels

High efficiency

Today's Linear Generator products deliver fuel to electrical efficiency of 45% or more, similar to fuel cells at full load.

Low emissions

Linear Generators demonstrate very low emissions of pollutants such as NOx, even when using hydrogen or biofuels.

Fuel flexibility

Can generate clean power from renewable fuels such as hydrogen, biogas, syngas, methanol and ethanol.

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Can generate clean power from renewable fuels such as hydrogen, biogas, syngas, methanol and ethanol.

..to meet rising global demand for responsive, cost-effective electrical power

Fast response

Can ramp up and down quickly to meet power load demands at a fraction of the emissions of reciprocating engine technologies.

Low cost

Best-in-class total cost of ownership through efficiency. Few moving parts and oil-free for low maintenance costs and long life.

Modularity

With no crankshaft, multiple cylinders in modular configurations can support operations of any size with cost-effective solutions

Libertine's modular solution

Cell

Libertine's core hardware technology

- ✓ Fuel-cell levels of efficiency for low fuel cost
- ✓ No efficiency compromise for variable fuels
- ✓ Remote diagnostics for low maintenance costs

➡ Lower \$/kWh operating cost

IGN60-P1 Linear Power System cell
60 kWe (First generation)



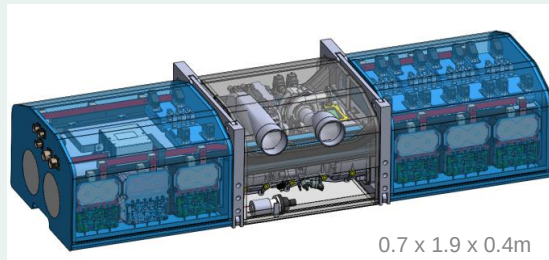
Module

Two-cell module for pack integration

- ✓ Integrates power electronics and controls
- ✓ Defined interfaces for pack auxiliaries
- ✓ Common module suits many pack applications

➡ Lower pack development cost

LGN120-P2 Linear Generator module
160-180 kWe (Second generation)



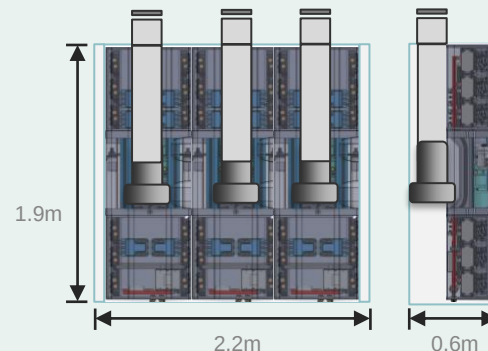
Pack

Module & aux package for product integration

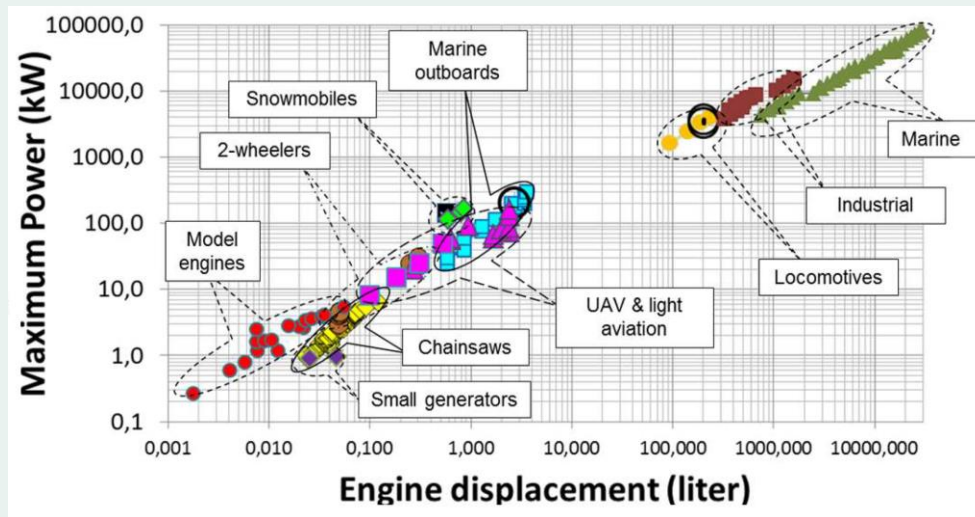
- ✓ Factory-configurable pack units for low installation cost
- ✓ Multiple modules per pack for high power density
- ✓ Shared power conversion and auxiliary services

➡ Lower \$/kWe CAPEX cost

iPU500-P2 pack unit
0.5 MWe (3x second generation modules)

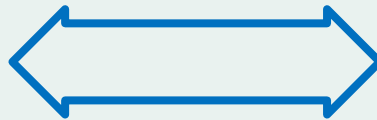


Linear Generator full potential 'sweet spot'

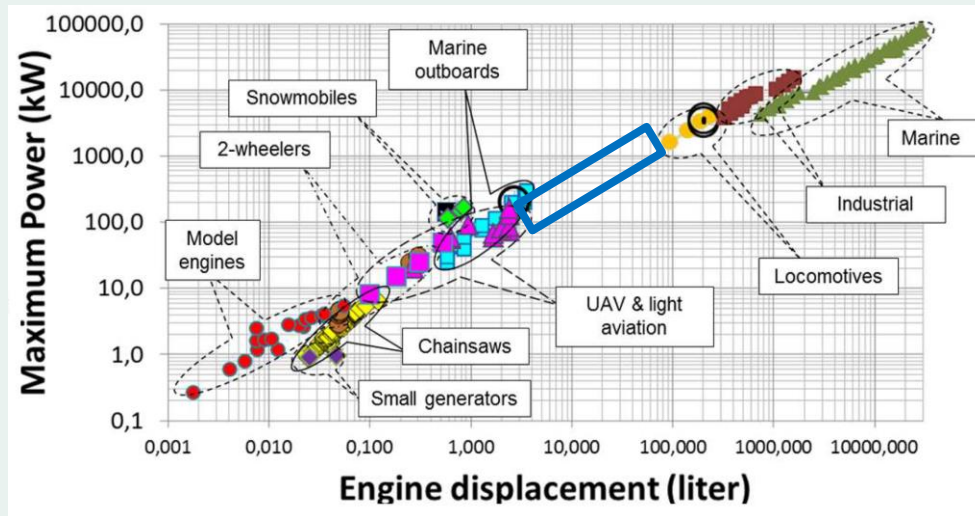


Linear Generator full potential 'sweet spot'

Packaged single cylinder
2.5L displacement
90kW max power



20' ISO packaged 36x cylinder
90L displacement
3000kW max power

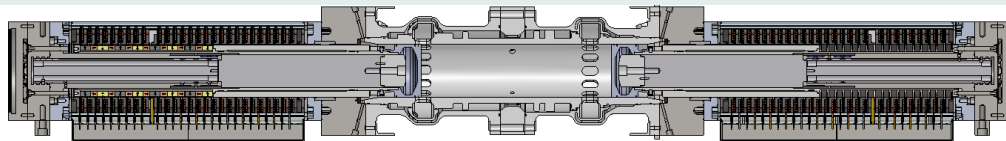


LGN60-P1 control & efficiency objectives

	Control KPIs			Efficiency KPIs		Durability KPIs	Operating range
	Compression ratio CR CoV%	Control margin vs. current limit	CR trend/ 1000 cycles	Indicated thermal efficiency	Net electrical efficiency	Max continuous run time	Compression ratio, fuel
LGN20-MVP achieved (2021)	1.04%	15%	N/A	37.6%	N/A	10 sec	CR 9 - 18, EtOH
LGN60-P1.4 target (2024-25)	< 1%	> 10%	<1 CR	55%	>40%	1 h	CR 9 – 24 EtOH, H2, CH4
Production target (2026-27+)	< 1%	> 10%	<1 CR	65%	>50%	8000 h	OEM choice

Key enabling features of LGN60-P1 vs. LGN20-MVP

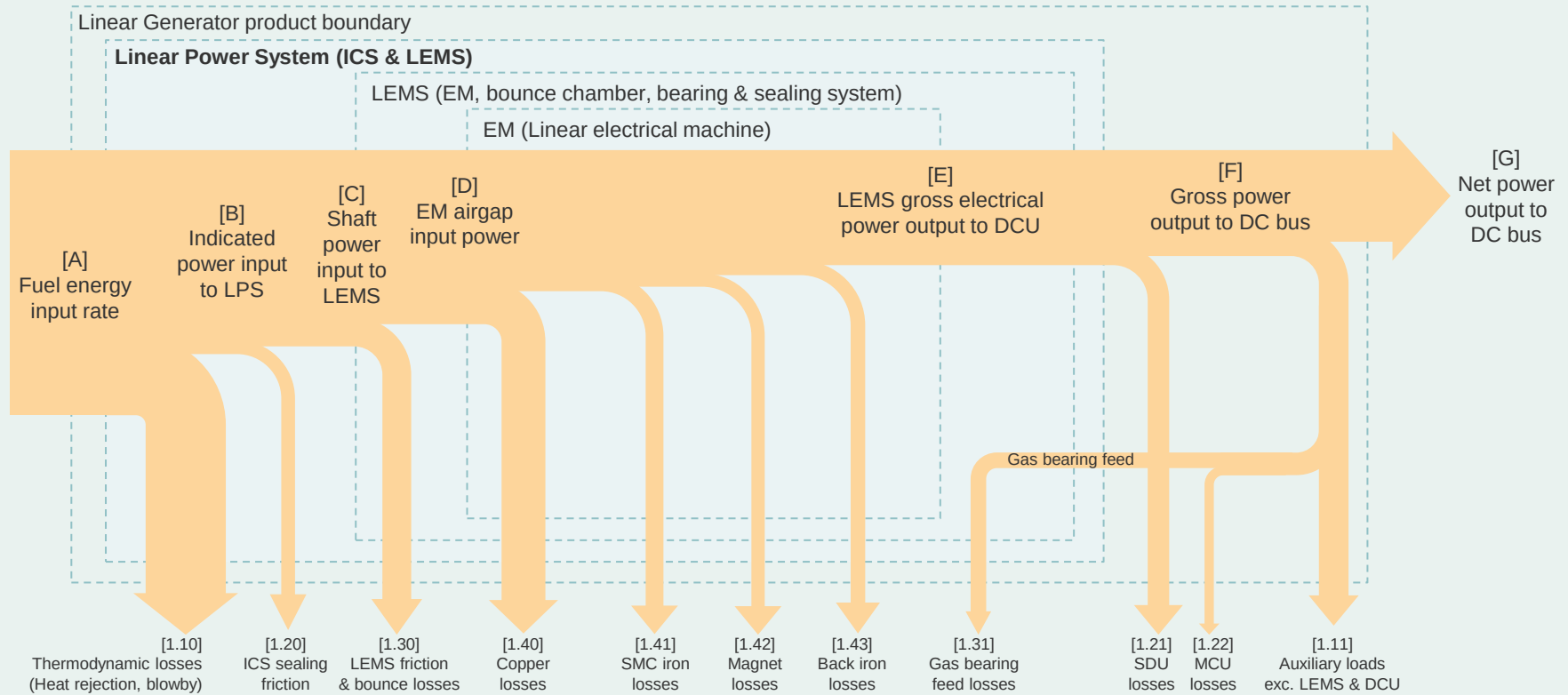
- High efficiency electrical machine
- High efficiency bounce chambers
- Internal gas bearings
- Larger bore & displacement
- Improved ports & scavenging
- Updated motion control software



System boundaries & energy flow definitions

Combustion operation

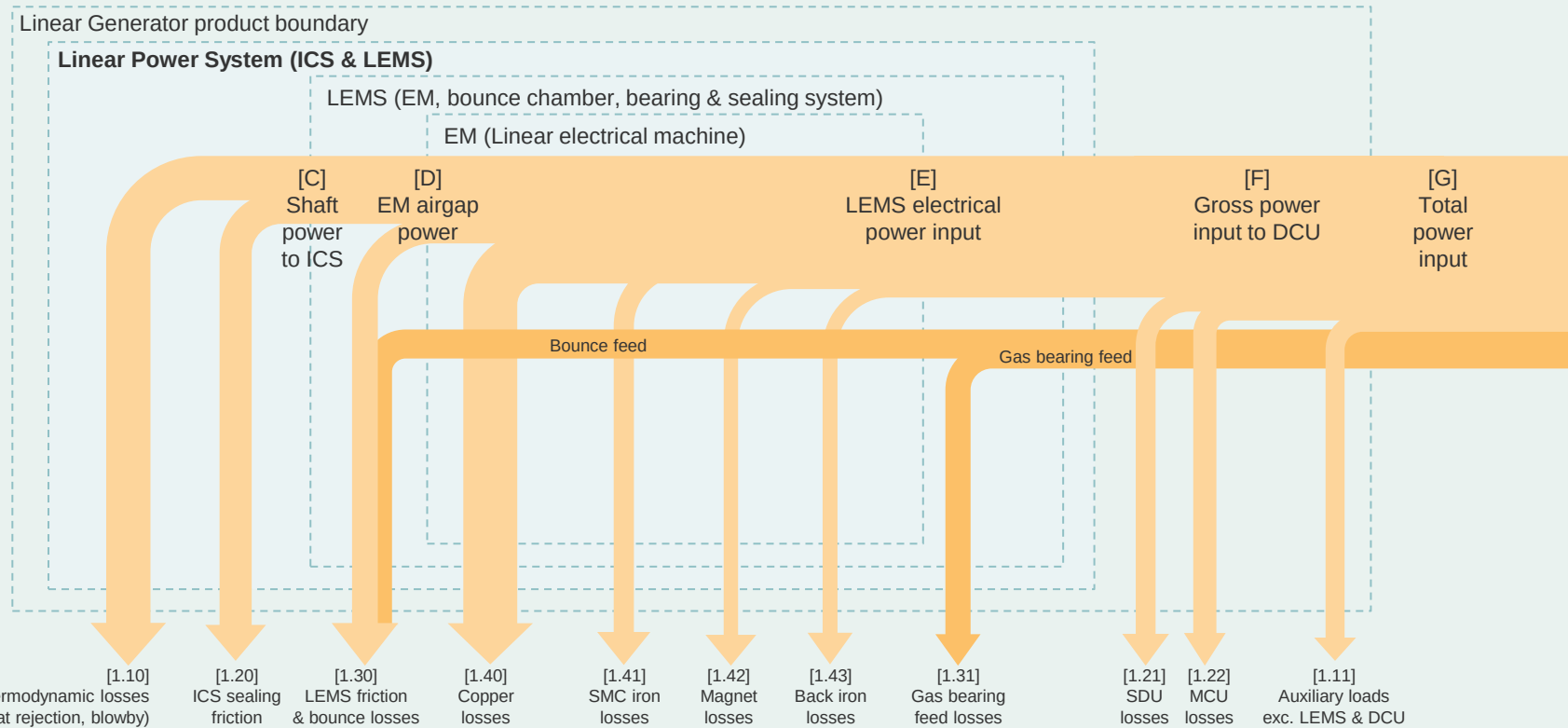
$$\text{Indicated thermal efficiency} = [B] / [A]$$
$$\text{Fuel LHV to DC electrical efficiency} = [F] / [A]$$
$$\text{Net electrical efficiency} = [G] / [A]$$



System boundaries & energy flow definitions

Motored operation

Indicated thermal efficiency = $[B] / [A]$
Fuel LHV to DC electrical efficiency = $[F] / [A]$
Net electrical efficiency = $[G] / [A]$



Efficiency performance (Combustion operation PROJECTED, LGN60-P1)

Energy balance illustration (J/cycle)

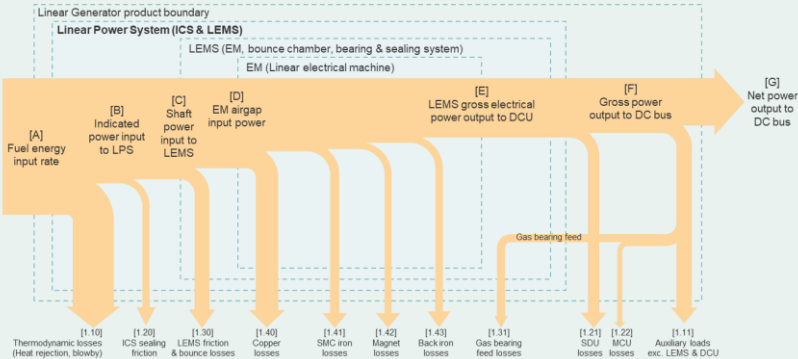
LGN60-P1.4 development cylinder, 2 movers & ICS

	Flow ID	Loss ID	Input	Losses	Net	Output	Notes
Fuel in	[A]		8418				
Thermodynamic losses (Heat rejection, blowby)		[1.10]		3788			Assume 55% indicated thermal efficiency
Indicated power to LPS	[B]				4630		
ICS sealing & friction		[1.20]		35			50N typical ring-pack friction
Shaft input to LEMS	[C]				4595		
LEMS friction losses		[1.301]		49			70N typical friction & hysteresis
LEMS bounce chamber losses		[1.302]		102			4.5% of stored energy
EM airgap	[D]				4444		
LEMS copper & iron losses (Nominal QFF current)		[1.4]		533			12% of input
LEMS copper & iron losses (PID error current)		[1.4]		89			2% of input
LEMS electrical power (energy)	[E]				3822		
SDU power electronics losses		[1.21]		222			5% of throughput
Gross electrical power (energy) to/from DCU	[F]				3599		
Bounce feed							TBD from flow at feed pressure
eBoost & bearing feed aux		[1.22]		200			Estimate
MCU supply & losses		[1.11]		20			Estimate
Net power (energy) output to DC bus	[G]					3379	
			100%	60%		40%	

Energy balance illustration (kW)

LGN60-P1.4 development engine, 2 movers & ICS

	Flow ID	Loss ID	Input	Losses	Net	Output	Notes
Fuel in	[A]		150				
Thermodynamic losses (Heat rejection, blowby)		[1.10]		67			Assume 55% ITE
Indicated power to LPS	[B]				82		
ICS sealing & friction		[1.20]		0.6			50N typical ring-pack friction
Shaft input to LEMS	[C]				82		
LEMS friction losses		[1.301]		0.9			70N typical friction & hysteresis
LEMS bounce chamber losses		[1.302]		1.8			4.5% of stored energy
EM airgap	[D]				79		
LEMS copper & iron losses (Nominal QFF current)		[1.4]		9.5			12% of input
LEMS copper & iron losses (PID error current)		[1.4]		1.6			2% of input
LEMS electrical power (energy)	[E]				68		
SDU power electronics losses		[1.21]		4.0			5% of throughput
Gross electrical power (energy) to/from DCU	[F]				64		
Bounce feed			0.3				TBD from flow at feed pressure
eBoost & bearing feed aux		[1.22]		3.6			Estimate
MCU supply & losses		[1.11]		0.4			Estimate
Net power (energy) output to DC bus	[G]					60	
			100%	60%		40%	



These loss components can be assessed and developed through motored testing

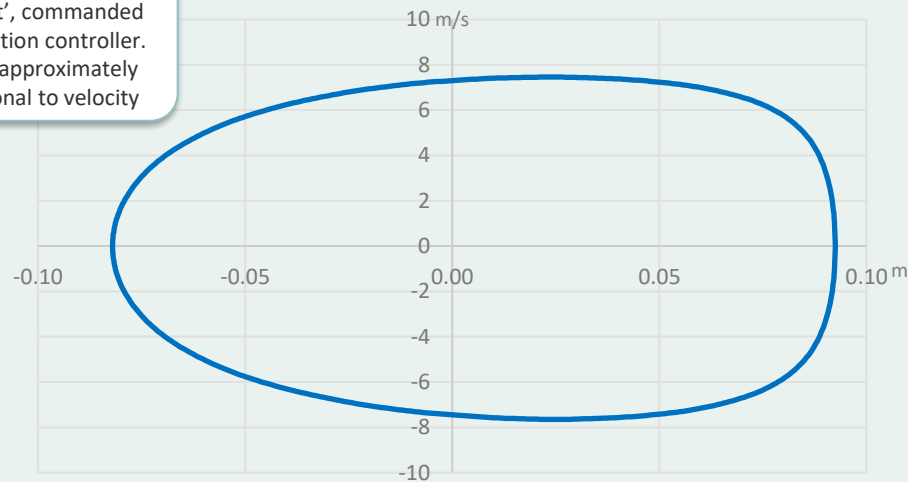
intelliGEN uses Q current 'feed forward' terms to minimise position, velocity and CR control errors

Phase current target, CR18 (motored) CLOPS71
(Commutated from QFF feed forward terms, positive stroke)



Phase current envelope is 'Q current', commanded by the motion controller. Target is approximately proportional to velocity

Target velocity profile, CR18 (motored) CLOPS71



Copper loss as a percentage of useful EM power recovered can be large, especially at low speeds

- Electromagnetic force function (EMFF) is approximately proportional to Q current

$$\begin{aligned}\text{Instantaneous useful power recovered} &= \text{velocity} \cdot \text{EMFF} \\ &= \text{velocity} \cdot Q \cdot k_1\end{aligned}$$

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- Copper losses are approximately proportional to the square of Q current

$$\begin{aligned}\text{Instantaneous copper loss power} &= Q^2 \cdot k_2 \\ \text{Copper loss / useful power} &= (Q / \text{velocity}) \cdot (k_2/k_1)\end{aligned}$$

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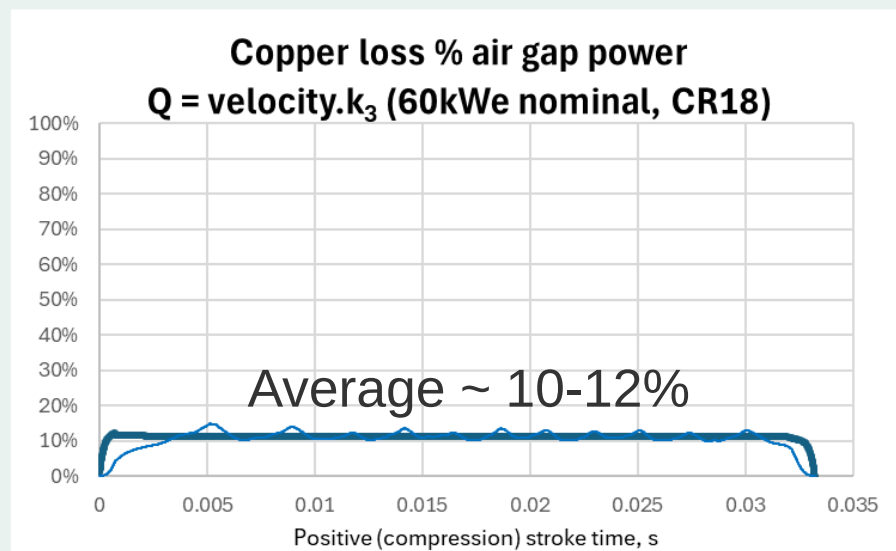
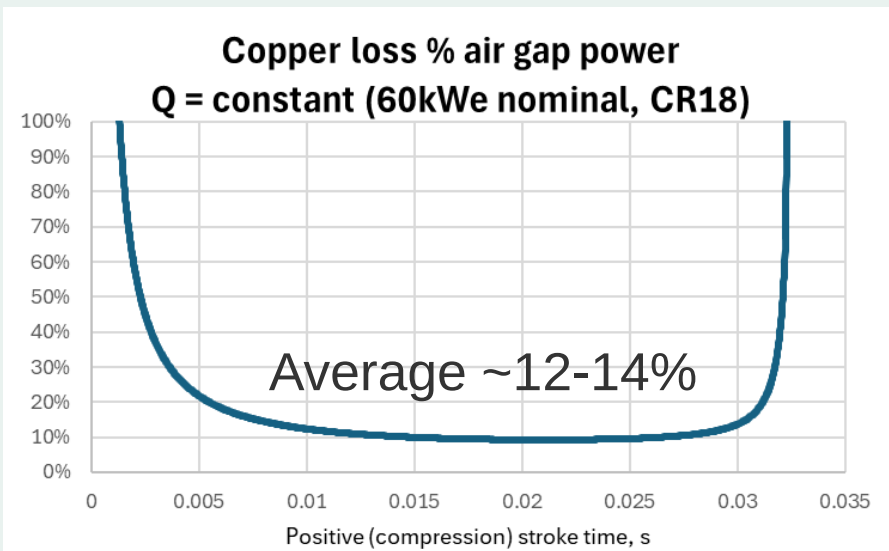
$$\text{Instantaneous copper loss power} = Q^2 \cdot k_2$$

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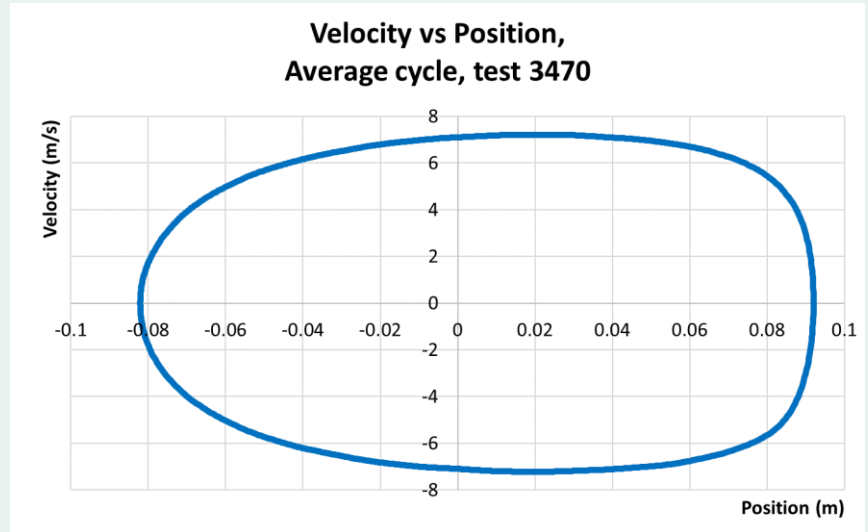
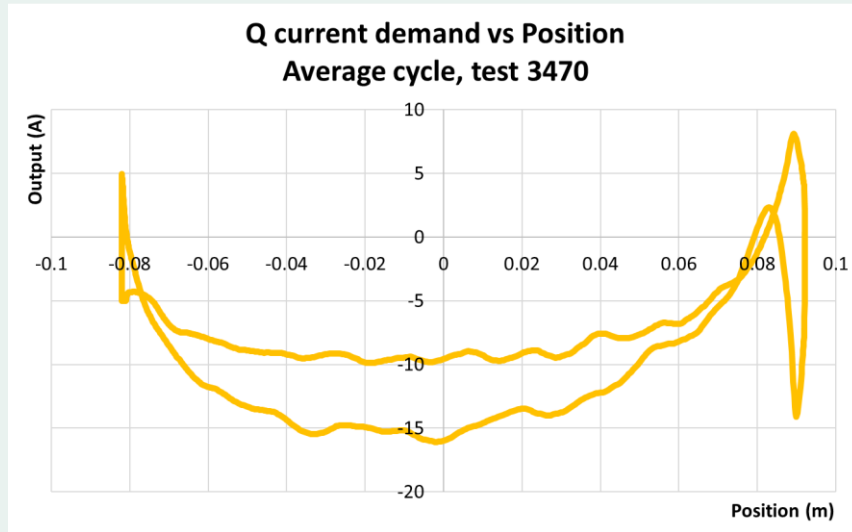
**Instantaneous copper loss %
is proportional to Q / velocity**

Copper loss as a percentage of useful EM power recovered can be large, especially at low speeds



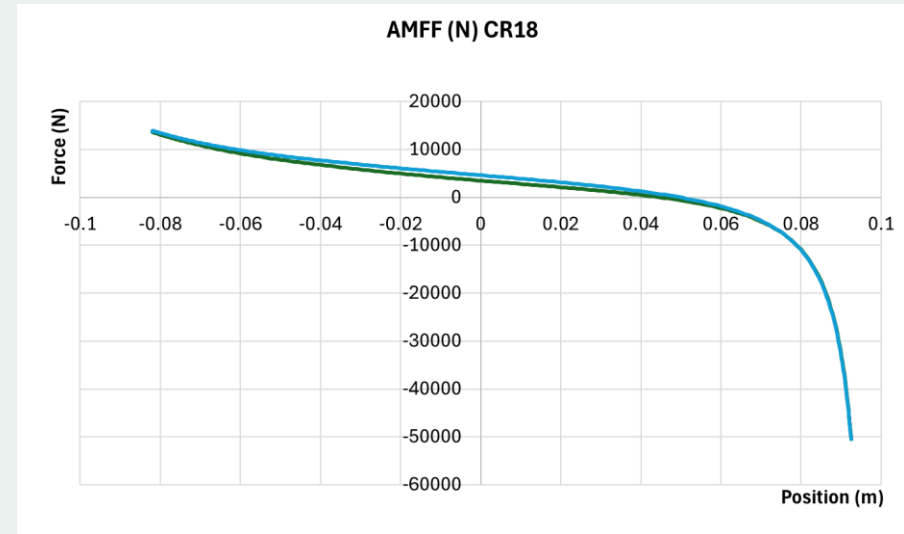
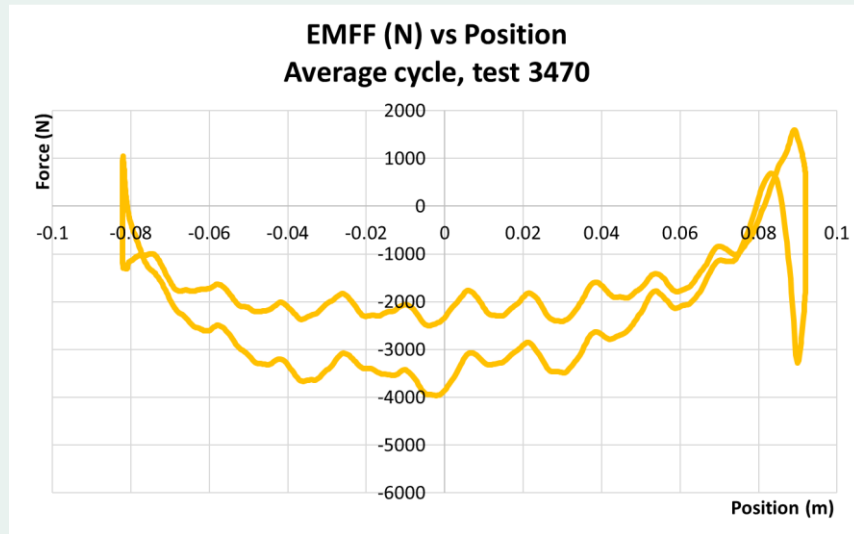
Control Library Operating Parameter Set (CLOPS) Q current & profile iteration method

1) Calculate average Q current demand and cycle velocity vs. position



Control Library Operating Parameter Set (CLOPS) Q current & profile iteration method

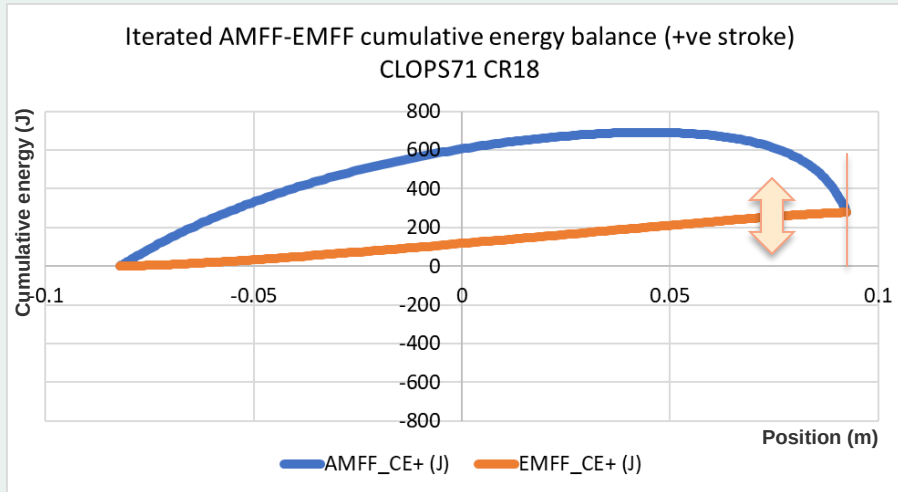
2) Calculate force functions for EM force (EMFF) and all other forces (AMFF)



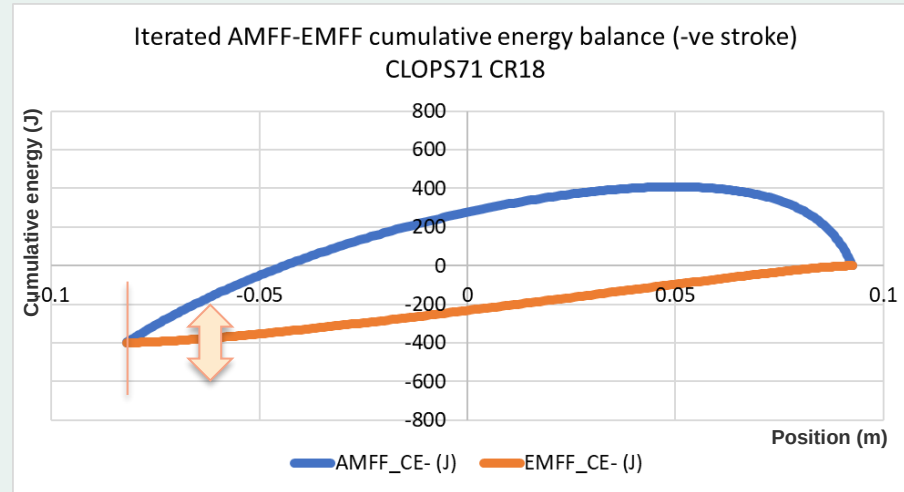
Control Library Operating Parameter Set (CLOPS) Q current & profile iteration method

3) Iterate k_3 (QFF/velocity) so that cumulative energy contributed by AMFF = cumulative energy absorbed by EMFF at the target end of stroke position

$$k_3 = \text{QFF/Velocity}^+ = -1.10 \text{ A/m/s}$$

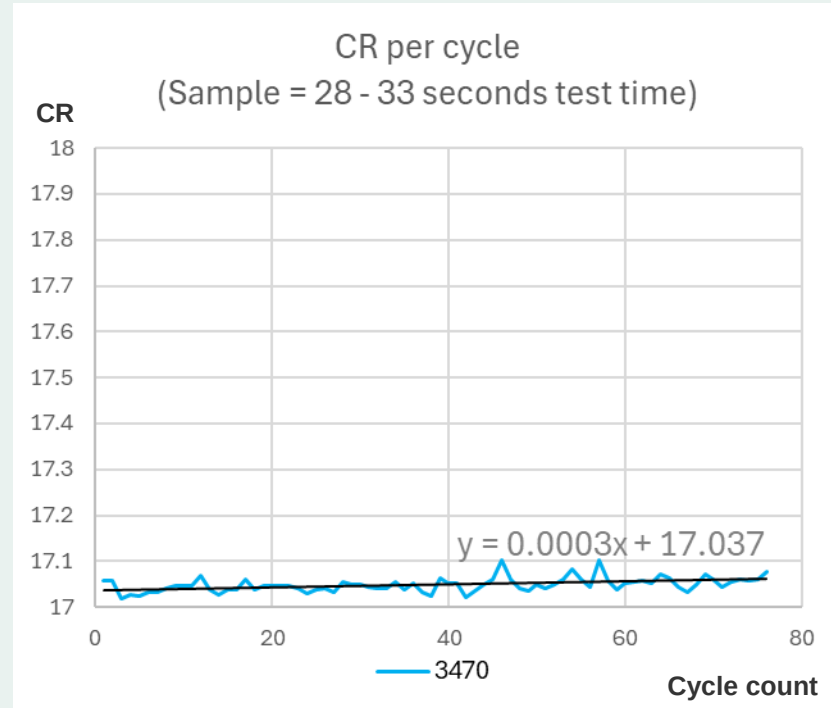


$$k_3 = \text{QFF/Velocity}^- = 1.55 \text{ A/m/s}$$



Empirical results, before iteration

Test 3470, CLOPS66, CR18 target



- Low CR variability (CoV ~0.1%)
- CR trend +0.3/1000 cycles as bounce chambers approach equilibrium
- CR17.04 achieved is below target CR18

Empirical results, before iteration

Test 3470, CLOPS66, CR18 target

LOG2024-KPI-001

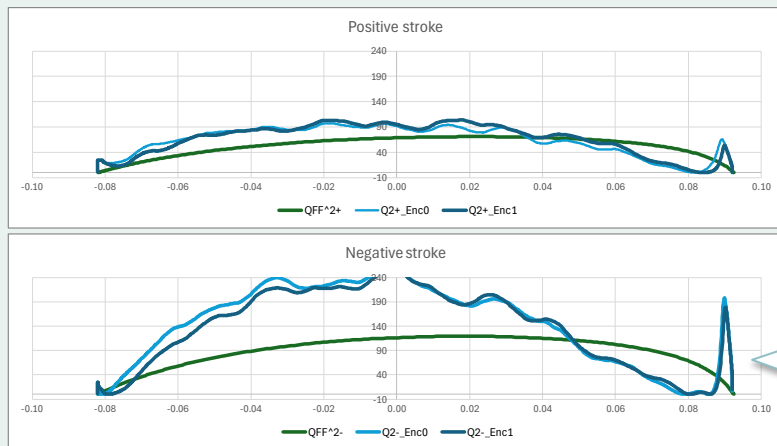
PID loss current performance

Test	LDT-MCU1-3470
Type	LPS HARDWARE TEST
H/L model ID	N/A
Test/H/L operator	Farouk Maloudi
CLOPS	CLOPS-IGN-2024.0062
CP1=	10
CP2=	18

Adapt script ID (rLAT)	N/A
Adapt script parameters (LAPS)	N/A
Script processed by (Name)	Jim Keown
Processing script ID	DCT2024-04-002-02B
Script time end	28
Script time start	33
# cycles	76.33

Positive stroke		Negative stroke	
Enc0	Enc1	Enc0	Enc1
42.5	42.5	73.8	73.8
56.4	57.0	124.1	116.6
14.0	14.5	50.4	42.8
33%	34%	68%	58%

Average (QFF^2) QFF2a
Average (Output^2) Q2a
PID loss current^2 PIDQ2a
PID loss current^2 % of QFF2a PID%QFF2a



- QFF current is proportional to velocity
- Motored performance shows actual Q current variations due to PID action
- Impact on copper losses is assessed as 'PID loss current^2', the amount by which demand exceeds target Q current (squared, time-averaged basis)
- The goal of CLOPS iteration is to improve efficiency by reducing PID loss current^2

QFF² is shown by the green line. Blue lines shown motored actual Q current variations due to control errors (PID action). For efficient operation, blue lines should match green lines

Empirical results, after iteration

Test 3503, CLOPS71, CR18 target

LOG2024-KPI-001

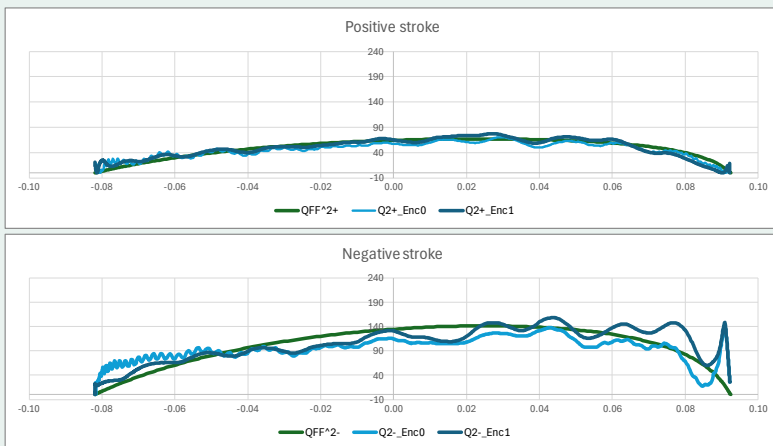
PID loss current performance

Test	LDT-MCU1-3503
Type	HARDWARE TEST
Build ID: HIL/LPS	IGN60-P1_003.B102
Test/HIL operator	
CLOPS	CLOPS-IGN-2024.0071
CP1=	0
CP2=	18

Adapt script ID (rLAT)	N/A
Adapt script parameters (LAPS)	N/A
Script processed by (Name)	TF
Processing script ID	DCT2024-04-002-02B.EXE
Script time end	38
Script time start	43
# cycles	77.81

Positive stroke		Negative stroke	
Enc0	Enc1	Enc0	Enc1
40.5	40.5	83.7	83.7
39.8	42.5	85.7	91.8
-0.7	2.0	2.0	8.2
-2%	5%	2%	10%

Average (QFF^2) QFF2a
 Average (Output^2) Q2a
 PID loss current^2 PIDQ2a
 PID loss current^2 % of QFF2a PID%QFF2a

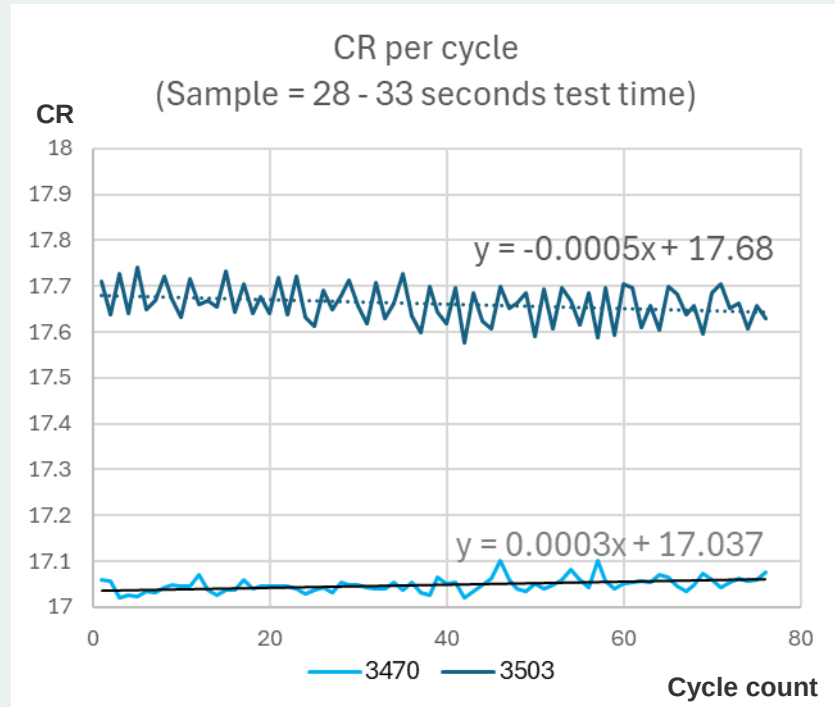


- CLOPS iteration reduces RMS current and PID loss current^2 , improving efficiency
- Additional benefit is reduced position, velocity and CR control errors
- In the motored CR18 target examples shown:

	CLOPS66	CLOPS71
Copper losses added by PID control variations (Enc0)	51%	0.3%
Copper losses added by PID control variations (Enc1)	46%	7%
Compression ratio error (mean)	-1.0 (17.0)	-0.3 (17.7)
Compression ratio (CoV%)	0.1%	0.2%

Empirical results, before iteration

Test 3503, CLOPS71, CR18 target

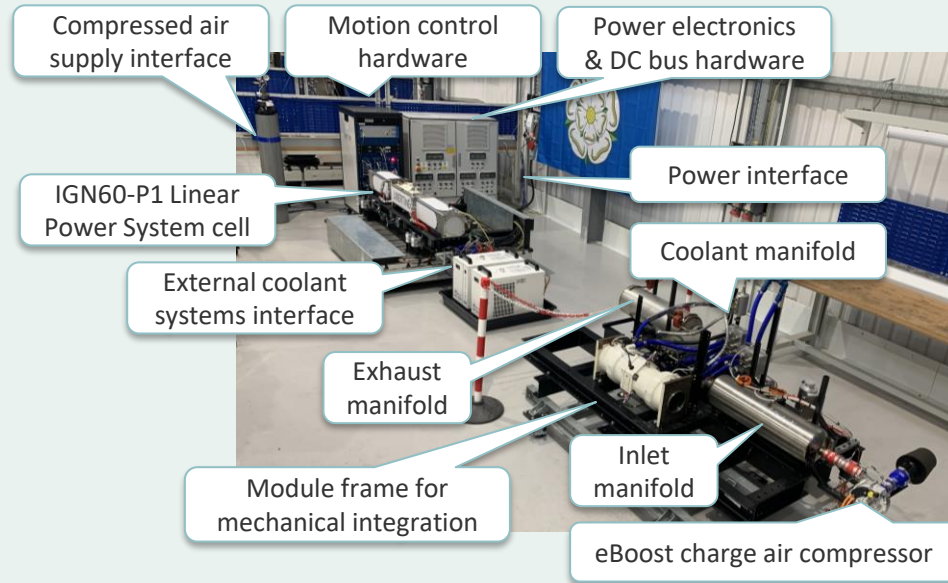


- CR17.68 achieved is closer to target CR18
- CR trend -0.5/1000 cycles
- CR shows cyclic variation, CoV 0.2%
- In the motored CR18 target examples shown:

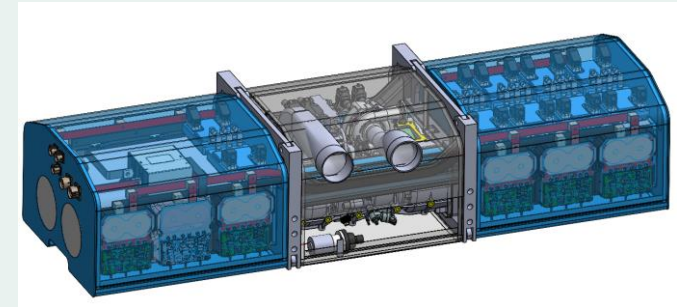
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P2 module integrates cells, auxiliary systems & interfaces into a compact, self-contained unit

LGN60-P1 Linear Generator & auxiliary systems
60 kWe (First generation)



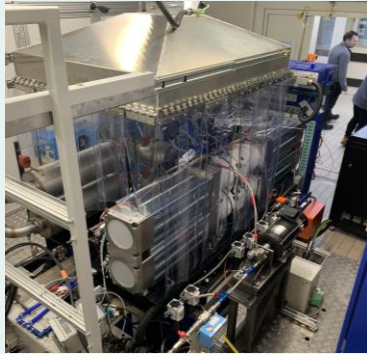
LGN120-P2 Linear Generator module
160-180 kWe (Second generation)



Libertine's customer propositions



P1 Product - For research organisations

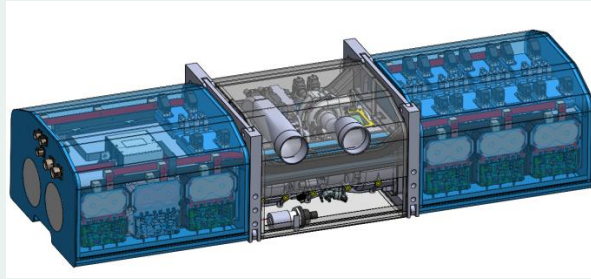


Opposed free piston research engine for advanced combustion and fuels

- ✓ Turn-key research engine solution with flexibility to reconfigure for alternative fuels and combustion systems
- ✓ Suite of tools for modelling, data acquisition and analytics for research productivity



P2 Module - For engine makers and powertrain businesses



Modular platform technology for Linear Generator product development

- ✓ Engineering services & prototype hardware for rapid performance validation and pilot demonstration
- ✓ Small batch and low volume module manufacturing
- ✓ Licensing for manufacturing scale-up and localisation



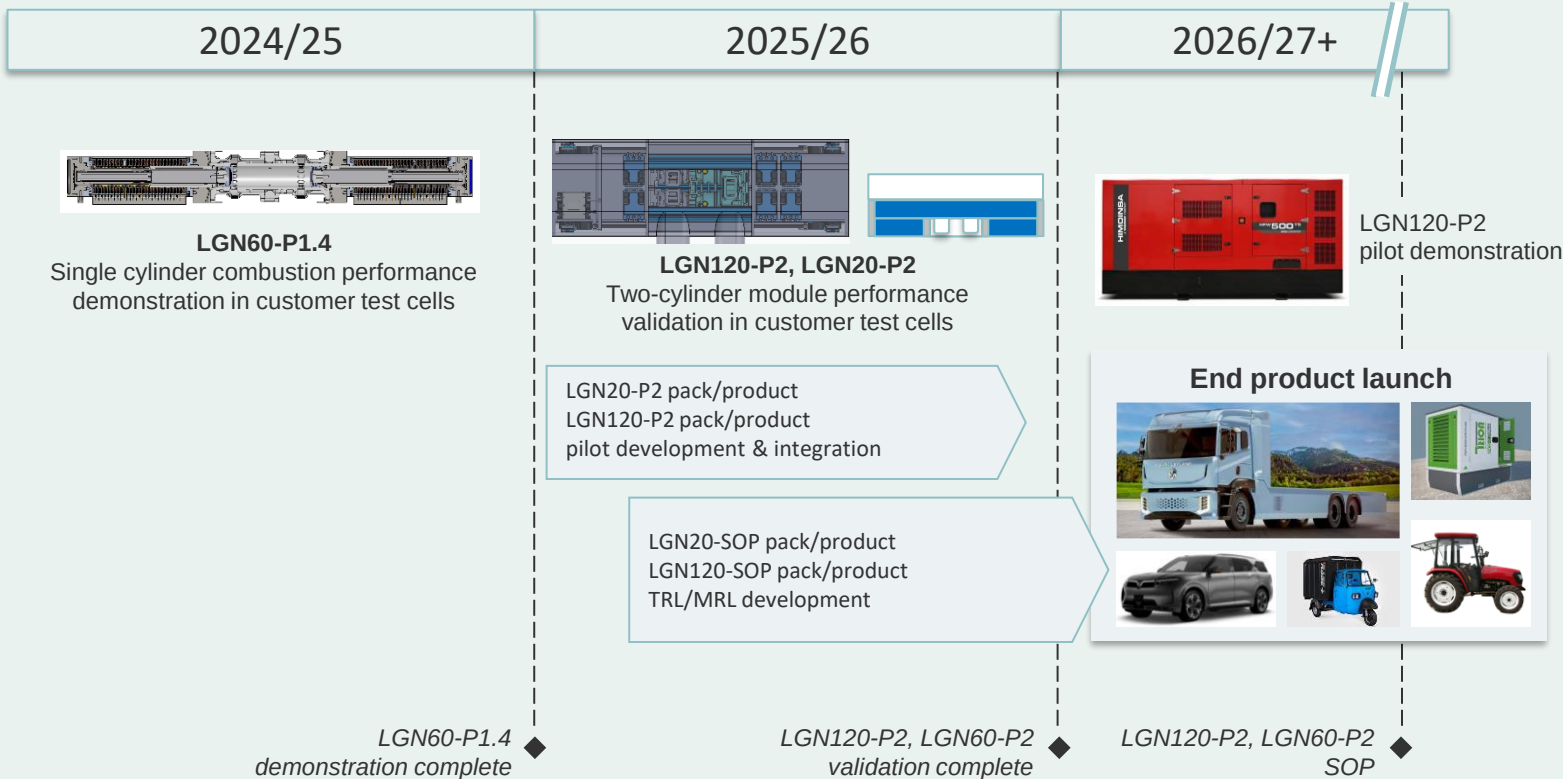
For heat engine and gas compressor businesses (non-core)



Hermetic linear motor-generator technology for power generation and other applications

- ✓ Engineering services & prototype hardware for rapid performance validation and pilot demonstration
- ✓ Licensing for manufacturing scale-up and localisation

New product development roadmap



Summary



A new power generator category for producing clean, efficient electrical power from a wide range of renewable fuels

Fuel-flexibility, modularity and form factor provide new opportunities for on-demand electrical power generation

May finally fill in the DI2S output/volume gap !

Precise, on-target compression ratio control with <1% CoV

Minimise linear e-machine losses, Q current demand proportional to velocity

Compact, high force linear electro-mechanical systems

'Next stroke control' to iterate 'Feed Forward' control current and target velocity profiles

Improved e-machine efficiency and control performance via Next-stroke control iteration, consistent with path to >50% fuel-to-electrical efficiency

OPEN FPE
LINEAR GENERATORS

Customer-configurable
Free Piston research engine systems

intelli GEN
LINEAR GENERATORS

Linear Generator modules & developer tools to accelerate pilot demonstration and new product introductions



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Thank you, & questions



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Austria

www.di2-stroke-engine-di2s.com

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